VitaLITy 2: A Study on LLM Usage in Literature Research

Yuhan Guo, Aiden Min, Hongye An, Yang Zhang, Jonathan Wang, Emily Wall, Arpit Narechania, and Kai Xu

Abstract—The rapid growth of scientific literature has made it increasingly challenging for researchers to efficiently identify, understand, and synthesize prior work. While recent advances in Large Language Models (LLMs) offer new opportunities to support literature research through semantic search, summarization, and natural language interaction, little is known about how these tools are actually used in practice or how they reshape research workflows. To understand LLM patterns of reliance in literature research, we conducted a qualitative user study with 12 visualization researchers. To facilitate our study, we built a system, VITALITY 2, an LLM-powered visual analytics system for literature research that integrates semantic embeddings, retrieval-augmented generation (RAG), and interactive visualization. Participants in our study used VITALITY 2 to conduct literature reviews and develop research ideas. Through interaction logs and semi-structured interviews, we examine how participants balance traditional search, visualization-based exploration, and LLM-driven features such as conversational querying and automated summarization. Our findings highlight that LLMs reshape literature research workflows by improving efficiency and facilitating hypothesis-driven exploration, yet critical tasks such as steering research directions, interpreting findings, and constructing knowledge remain human. We identify key opportunities for designing future LLM-supported literature research tools, particularly in integrating interactive visualization with AI-driven workflows to better support complex sensemaking tasks.

Index Terms—LLM, literature research, literature review, visualization

1 Introduction

With the accelerated pace of scientific discovery, it is becoming increasingly difficult to keep up with the growing body of literature. However, this is essential for not only keeping abreast of one's own field, but also when exploring a new area. This is particularly daunting for fast moving fields such as machine learning, with tens of thousands papers published each year.

There has been a long history of work targeting these challenges of literature research [1], with many of these tools employing Data Visualization, Natural Language Processing (NLP), or both. The advent of the transformer-based models (such as BERT [2] and SPECTER [3]) and more recently frontier Large-Language Models (LLMs) such as GPT and Gemini series, have completely transformed these tools. They dramatically improved the semantic capability of such systems with more advanced text embeddings that enabled the discovery of relevant publications through semantic similarity. They also completely changed how users interact with these tools: LLMs allow users to perform complex queries by prompting in nature language and enable new analyses that were not possible before, such as summarizing a paper. Finally, agentic workflows built on top of LLMs make it possible to automate the entire literature research process, e.g., Deep Research modes in ChatGPT and Gemini.

These new capabilities enabled a new generation of literature research tools, with VITALITY [4] being one of the early examples. The system included a web-based visualization interface that used text embeddings to identify semantically similar literature based on input papers using SPECTER [3] and GloVe [5]. Since then, there have been many new research prototypes and studies on this topic such as OpenScholar [6] that adopted a Retrieval-Augmented Generation (RAG) approach [7] and the study by Delgado-Chaves et al. [8] that showed LLMs can significantly reduce paper screening efforts in literature research. There are also many commercial offerings emerging in this space, such as Consensus [9], Elicit [10], and Undermind [11].

However, in spite of the growth of tooling in this space, there is a sparsity of research on how people use these tools, how that changes as LLMs become more capable, or how this changes the way literature research is generally conducted. A notable exception is the study by Wang et al. [12] that investigated how early-stage researchers use LLM for literature research. While informative, the study mainly used an off-the-shelf chat interface and was based on now-outdated LLM models. To fill this research gap, we conducted a study to better understand how researchers are performing literature research with specialized tools, especially those with data visualization support, coupled with the latest LLMs and technologies such as RAG.

First, we want to make an important distinction between "literature search" and "literature research": the former is mostly about finding relevant publications, whereas the latter also covers understanding the technical details and synthesizing knowledge from multiple sources. Our study focuses on the latter, of which the former is a part.

As discussed earlier, we would like to find out:

1. How researchers are using LLM-powered literature research tools? Do they change the way of conducting literature research?
2. What is still missing for AI-assisted literature research, especially from the interactive visualization perspective?

To answer these questions, we conducted a qualitative user study with N=12 participants, who were all experienced visualization researchers. During the study, each participant conducted in-depth literature research to identify a new question for the next research project in the area of their choice. The results show that LLMs transform literature research workflows by improving efficiency and reducing cognitive load while enabling more top-down, hypothesis-driven approaches, yet critical tasks such as steering research directions, interpreting findings, and knowledge construction remain inherently human activities. We identify key opportunities for visualization to enhance validation, explainability, and collaborative knowledge construction in AI-augmented literature research.

To support the study, we developed VITALITY 2, an open-source visualization system for literature research. To ensure the system reflects the start of the art, it is based on the latest LLM (GPT 5.4), allows complex query through prompting, and incorporates Retrieval Augmented Generation (RAG) architecture and agentic workflows optimized for literature research. VITALITY 2 is not just designed for

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this study: it is intended to be a fully-fledged online system serving the visualization research community. The system contains the information of over 75,000 publications from 38 visualization and related conferences and journals, covering the period from records began to late 2025/early 2026. This makes it one of the largest public paper collections in the field of visualization. New publications will be added as they become available. VITALITY 2 is available online at https://vitality.mathcs.emory.edu/vitality2/ and free for everyone to use.

In summary, our research contributions include:

1. An empirical study that provides new insights on how LLM-powered tools are being used for literature research and how the latest AI features are changing researchers' behavior,
2. Identification and synthesis of research opportunities for future LLM-based literature research tools, and
3. A state-of-the-art LLM/RAG-powered literature research tool that is publicly available and tailored for the visualization community.

2 Related Work

2.1 LLMs for Literature Research

Natural Language Processing (NLP) methods have long been used to support literature research. Before the arrival of the transformer architecture [13], such tools often relied on techniques such as Latent Dirichlet Allocation (LDA) [14] and text embedding methods such as Word2Vec [15] and GloVe [5]. Here we focus on the efforts after the introduction of transformer models, which are also commonly known as Large-Language Models (LLMs). BERT [2] is probably the best know early LLM. SciBERT [16], based on BERT, is trained specifically for scientific text, and SPECTER [3] is a further improvement by including citation information, and which targets document-level embedding. The best known literature research tool based on this generation of LLMs is probably Semantic Scholar [17], providing semantic-similarity based *literature search* (not *literature research*).

Modern LLMs, also known as frontier models, further increase the semantic capability of these tools dramatically. Even without specialization, such models (e.g., the GPT and the Gemini series) can already perform simple tasks during literature research, such as publication screening [8], and have been widely used in practice [18]. However, there are two potential weaknesses of using LLMs directly: hallucination [19] and a lack of the latest information. The latter is the result of the training data cut-off date, after which the LLM has no information of anything after that, including the latest publications. One popular solution to both is Retrieval Augmented Generation (RAG) [7], which employs information retrieval methods to find relevant and more up-to-date information to include in prompts. For example, OpenScholar [6], which uses a RAG architecture, outperforms general purpose LLM GPT-4o while being a much smaller model.

Another important change brought about by frontier models is agents enabled by them, which make it possible to fully automate the entire literature research process. Some of them specifically target literature research, such as AutoSurvey [20], while others are part of the larger automatic scientific discovery systems, such as the literature research component of the fully automated AI Scientist system [21]. However, such systems are not without their controversies. For instance, arXiv recently started to reject any non-peer-reviewed literature review or survey [22]. This also raises an interesting question: who is literature research for? For systems like AI Scientist, the output of literature research is mostly for the consumption of downstream agents. As a result, they have a very different set of requirements from those systems designed for humans. In our study, we focus on work targeting humans, i.e., tools that help users to better identify, understand, and synthesize knowledge from academic publications.

2.2 Visualization for Literature Research

Visualization methods for literature research are closely linked to those intended for text visualization [23]. For example, early text visualization work like ThemeView [24] can be easily applied to a publication corpus. Similarly, there are many contributions from outside the visualization and HCI communities (such as the work by El-Arini and Guestrin [25]). It is beyond the scope of this paper to review all such efforts, so we focus our discussion on those techniques targeting literature research specifically within the visualization and HCI communities.

Broadly speaking, the visualization research for literature analysis spans a spectrum, from providing an overview of a field to helping find and understand papers related to a specific topic. For the former, a well-known example is the work by Isenberg et al. [26] that aims to provide an overview of the entire visualization research landscape, using papers spanning two and a half decades. More recent examples include City of Wander [27] that deploys a cityscape metaphor to visualize over 20 years of TVCG publications, and HINTs [28] that uses LLM to better support sensemaking of large document collections by tailoring the visualization to users' analysis tasks, which can be easily extended to literature research. Related to these are efforts for knowledge visualization, such as the Atlas of Science [29] and the work by Chen [30].

The other end of the spectrum targets literature for a specific topic. Common approaches include topic or semantic analysis, network analysis, and supporting the literature review workflow. From early on, it has been recognized that keyword matching alone is not an effective way to find relevant papers [25, 31, 32], as a concept or technique can often be expressed in several different ways. To address this, many approaches use text analytics or natural language processing methods. Topic modeling is a popular approach among early work that can help identify relevant papers by grouping similar ones together [25, 31, 32]. This includes techniques that are designed for text analysis in general and can be easily applied to literature analysis such as Serendip [33]. As NLP research progresses, new techniques are being used for literature analysis, such as text embeddings generated by transformer-based models [34], including GloVe [5] and SPECTER [3] used in VITALITY [4]. While visualization methods are starting to take advantage of the embeddings afforded by frontier models (as discussed in Section 2.1), the focus has been on improving user experience, such as providing on-demand explanations and other support during reading [35].

Besides LLMs, visual tools have been designed to provide interactive assistance for various user tasks in literature research, with or without AI support. As discussed in Section 2.1, while it is possible to fully automate literature research, here we focus on efforts targeting humans rather than agents. The work by Ma et al. [36] allows users to organize papers for literature review through sketching. LitWeaver [37] uses LLMs to scaffold writing during literature research. CrossLit [38] provides a more integrated environment that allows smoother transitions between the information foraging and writing stages common in literature research. Finally, systems such as Relatedly [39] and Dimind [40] target the entire literature review workflow.

Additional user information is often used to help uncover relevant literature. CiteSee [41] utilizes users' reading history to help prioritize new literature discovery. Following a similar approach, PaperWeaver [42] uses information from already collected papers to add context to help users better understand the relevance of new recommendations. Combined with the embedding approach, recent work such as VADIS [43] introduces a dynamic and personalized embeddings to better meet user preferences and improve transparency.

Network analysis is another common approach used by literature visualization, creating and visualizing co-author and affiliation networks [31], citation networks [44–46], and similarity networks [4, 26, 34, 44], among others. PUREsuggest [47] uses the information from the citation network among user-selected publications to further improve recommendations, which also allows user steering through keywords. Finally, Litforager [48] utilizes the immersive characteristics of VR to support literature network exploration.

2.3 Empirical Studies on Literature Research

With the increasing popularity of AI tools, there is a growing body of studies looking into how this trend changes the way we conduct research and its impact on users. These studies cover a range of topics, such as the impact on academic writing [49–52], critical thinking [53, 54], and HCI research in general [55]. Herein, our focus is work related to literature research, especially in the context of generative AI.

There is surprisingly limited research on this topic. The main one we found is the user study by Wang et al. [12] on people using LLMs for literature research. It targets exclusively early-stage researchers and found that they are generally open to LLMs and can adapt them for the purpose of literature research. While informative, the study mainly used an off-the-shelf chat interface and was based on now-outdated LLMs. Our study aims to better understand how specialized features, especially data visualization, affect LLM-supported literature research. We also try address whether improved capabilities of the latest LLMs and new technologies such as RAG and agentic workflows change users' behavior. Finally, we want to identify any remaining obstacles and future research questions for this topic. All of these are missing from the current literature.

In the context of visualization research, the study by Choe et al. [56] tries to understand how people understand citation network visualization, though this is not AI related. A recent case study by Wang et al. [57] demonstrated that LLMs can be effective in analyzing visualizations from application domains for literature research, but human oversight is essential to achieve the best results.

3 VITALITY 2

In this section, we describe the architecture, implementation, and user interface design of VITALITY 2, an LLM-powered visual analytics tool to help users perform and draft academic literature reviews. We will also utilize this system as our user study prototype.

3.1 Architecture and Implementation

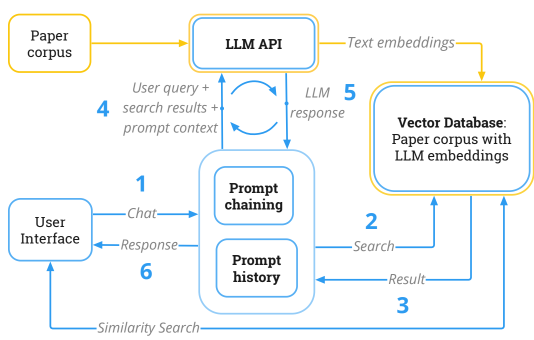

Fig. 1: Architecture of VITALITY 2: **(Step 1)** User input to the system. **(Step 2 & 3)** Retrieve data from the vector database. **(Step 4)** Combine the result with user input in the prompt. **(Step 5)** Recall result from LLM. **(Step 6)** Return the final result to the user.

Figure 1 illustrates the architecture, flow of data, and core links of VITALITY 2. All the papers in the corpus are first pre-processed using an LLM to create their embeddings (for similarity search), with the results stored in a vector database. This is shown as the orange part on the top. Similarity search is then performed within the vector database without further LLM access (the bottom line). User input from the chat interface (Step 1) is broken down into a series of smaller tasks using "Prompt chaining." For each sub-task, relevant information is retrieved from the vector database (Step 2 & 3) and then combined with user input in the prompt before sending it to the LLM (Step 4). "Prompt history" provides context from previous conversations. Once all the sub-tasks are completed (Step 5), the final results are returned to the user (Step 6).

3.1.1 Similarity Search using LLM

All the papers in the corpus are pre-processed using an LLM to create their embeddings based on paper metadata, which includes the title, authors, the conference or journal in which the paper is published, publication date, keywords, and abstract. These text embeddings convert textual data into high-dimensional vectors that capture semantic relationships among paper metadata [58], ensuring thorough comprehension of each paper and enabling more effective similarity searches within the vector database. For clarity, the "text embedding" mentioned refers to embeddings created from paper metadata rather than the full paper content. This metadata-focused approach balances detail with computational efficiency and allows for robust semantic matching.

To find similar papers from an academic literature corpus, while VITALITY uses GloVe [5] and SPECTER [3] embeddings, VITALITY 2 uses ADA, the embeddings generated by OpenAI's *text-embedding-ada-020* model [59], which is a breakthrough model that generates contextually rich embeddings by leveraging advanced attention mechanisms and transformer architectures. Thus, VITALITY 2 now includes ADA, GloVe, and SPECTER embeddings. VITALITY 2 also leverages two *vector databases* (Faiss [60] and ChromaDB [61]) to store and manage these embeddings and also to locate similar papers via approximate nearest neighbor search.

3.1.2 Chat with Papers using RAG and Prompt Chaining

While text embeddings and vector databases allow finding similar papers, there are several other kinds of analyses a user may want to perform, such as understanding a technical concept, summarizing a single paper, or writing a literature review based on multiple papers. To perform such analyses, users may need to build customized tools and/or learn a new query language, neither of which are readily accessible to nor understandable by many users. VITALITY 2 addresses this issue by leveraging the powerful natural language understanding and generation capabilities of LLMs, allowing users to directly engage with the LLM using natural language.

However, there are a few challenges when applying LLMs for such analyses: (1) *Prompt Size* – This refers to the largest prompt that an LLM can accept and depends on the LLM type and version. Earlier versions of LLMs accepted a few thousand tokens, wherein each word in a prompt accounted for a few tokens. More recent versions have a larger limit, e.g., up to 16K tokens for GPT 3.5 [62]. However, this can still limit analyses involving a large corpus of papers, which often have thousands of words each. (2) *Hallucination* – it is well known that LLMs can create seemingly plausible information about something that does not exist [63], e.g., suggest papers that never existed, that can be detrimental to reliable literature research. (3) *Limited Comprehension Capabilities* – Despite the ability to generate seemingly logical and coherent text, LLMs lack genuine comprehension capabilities. This implies that an LLM may encounter difficulties when handling complex problems or tasks, particularly those requiring a deeper understanding of context or concepts.

To overcome these issues, we employ two popular approaches: Retrieval Augmented Generation (RAG) and Prompt Chaining.

RAG. RAG semantically processes the user's input query to only retrieve relevant information from within the data corpus, thereby reducing the subsequent prompt size and also minimizing the risk of hallucination. This retrieval process is not about finding exact answers but rather fetching documents that contain potentially useful context or information related to the query.

Prompt Chaining. Prompt chaining breaks down complex tasks to a series of smaller steps and provides specific prompts that are known to be effective for each of these steps [64]. VITALITY 2 borrows this idea and implements a conversation framework to manage the content and context of user queries and LLM responses. VITALITY 2 uses LangChain [65] a popular open-source library for this purpose. As LLM API calls operate independently of each other without inherent memorization or management of the context, there is no built-in functionality for retaining conversation history. Sending the entire dialogue history with each LLM API call risks exceeding the maximum token limit. Therefore, we adopt a two-step approach in VITALITY 2: first, a summary of recent conversations is generated by calling the LLM API once to obtain a "condensed conversation history." Then, in the second API call, the user's query, conversation history, and retrieved information are concatenated into a prompt and sent to the LLM API. This is an example of prompt chaining that efficiently combines the results of two or more LLM API calls.

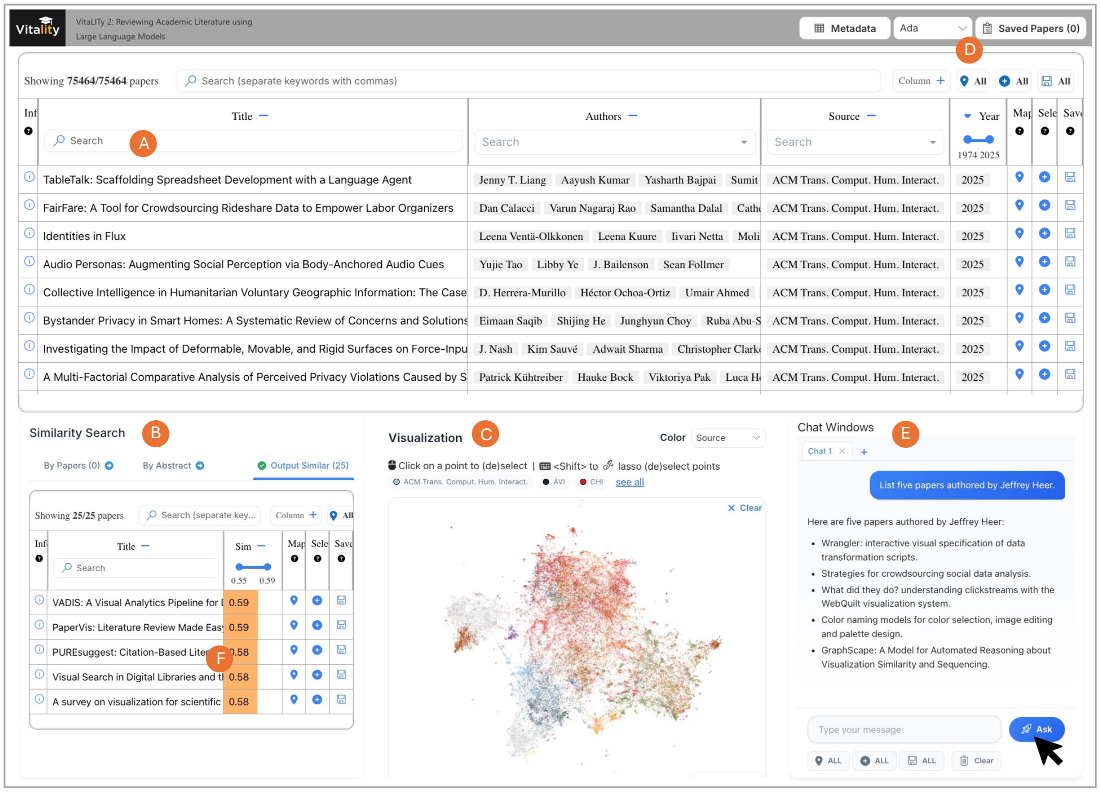

Fig. 2: The VITALITY 2 User Interface. (A) **Paper Collection View** shows the entire corpus of publications, (B) **Similarity Search View** shows options to look-up publications that are similar to another list of publications or by a work-in-progress title and abstract, (C) **Visualization Canvas** shows an interactive 2-D UMAP projection of the embedding space of the entire paper collection, (D) Opens a **Saved Papers View** from which the saved papers can be exported as JSON. Extending VITALITY, we added ADA embeddings (in addition to GloVe and SPECTER embeddings). We also added the (E) **Chat Windows** view to allow users to ask natural-language based questions based on the paper corpus.

3.2 Dataset of Academic Articles

VITALITY [4] provided a dataset of 59,232 academic papers from visualization and HCI literature, along with their metadata such as their title, abstract, author(s), keyword(s), publisher, publication year, citation counts, and n-dimensional and 2-dimensional vector embeddings (GloVe and SPECTER). VITALITY also open-sourced a web-scraping framework to extract the above information from digital repositories such as IEEE Xplore and ACM Digital Library for continued development of the corpus. VITALITY 2 extended this collection by adding more recent papers between 2021-2026 (partial) through Semantic Scholar API [66], resulting in an augmented dataset of 75,464 papers.

3.3 User Interface

We built VITALITY 2 by extending VITALITY, as shown in Figure 2 and Figure 3. In VITALITY, the **Paper Collection View** (A) shows the entire corpus of publications, the **Similarity Search View** (B) shows options to look-up publications that are similar to another list of publications or by a work-in-progress title and abstract, the **Visualization Canvas** (C) shows an interactive 2-D UMAP projection of the embedding space of the entire paper collection, and (D) opens a **Saved Papers View** from which the saved papers can be exported in a JSON and .bibtex format for later use. We added three new capabilities in VITALITY 2, detailed next.

First, we expanded the available embedding options (from GloVe and SPECTER) to add OpenAI's ADA [67]. Users can view the 2-dimensional ADA embeddings in the UMAP and search for similar papers (by title or abstract). Based on our own testing and consistent with prior benchmarks [68], we found ADA embeddings to perform better than GloVe and SPECTER on VITALITY 2 features.

Second, we added a new **Chat Windows View** (E) to allow users to ask natural-language based questions based on the entire corpus, e.g., *"List five papers authored by Jeffrey Heer."* For the papers mentioned in the LLM's response, VITALITY 2 allows the user to map (view in the UMAP), select (look up other similar papers), or save them (include it in a literature review or export). This capability holds immense potential for revolutionizing scholarly information access, enabling users to effectively harness the capabilities of RAG and pose a wide spectrum of inquiries.

Third, building upon the robust semantic understanding capabilities of LLMs, we also added the capability to utilize the saved papers (**LLM Context View**) and automatically generate a literature review in the **LLM Output View** (Fig. 3F). We also introduce a document canvas that allows users to draft and organize their own literature review (**Research Notes View**) directly within the system (G). The canvas supports plain-text editing along with standard formatting options. Clicking the "Literature Review" button outputs a comprehensive literature review based on the *LLM Context*. These enhancements enable users to swiftly grasp the key information contained within their saved papers or draft an early version of their related work section in their ongoing manuscript. Note that VITALITY 2 allows users to customize the base LLM prompts to control the verbosity, writing style, and format of the LLM's response.

4 METHOD

We present a qualitative study designed to understand how people conduct AI-assisted (visualization) literature research, using the VITALITY 2 system described in the previous section. We sought to understand (i) how participants use AI within the tool as opposed to non-AI features, (ii) their perceived quality of the output with and without AI, (iii) their perceived efficiency when using AI supported features, and (iv) any trust concerns such as hallucinations, reduced understanding of the topic, etc. Fig. 4 is an overview of the study.

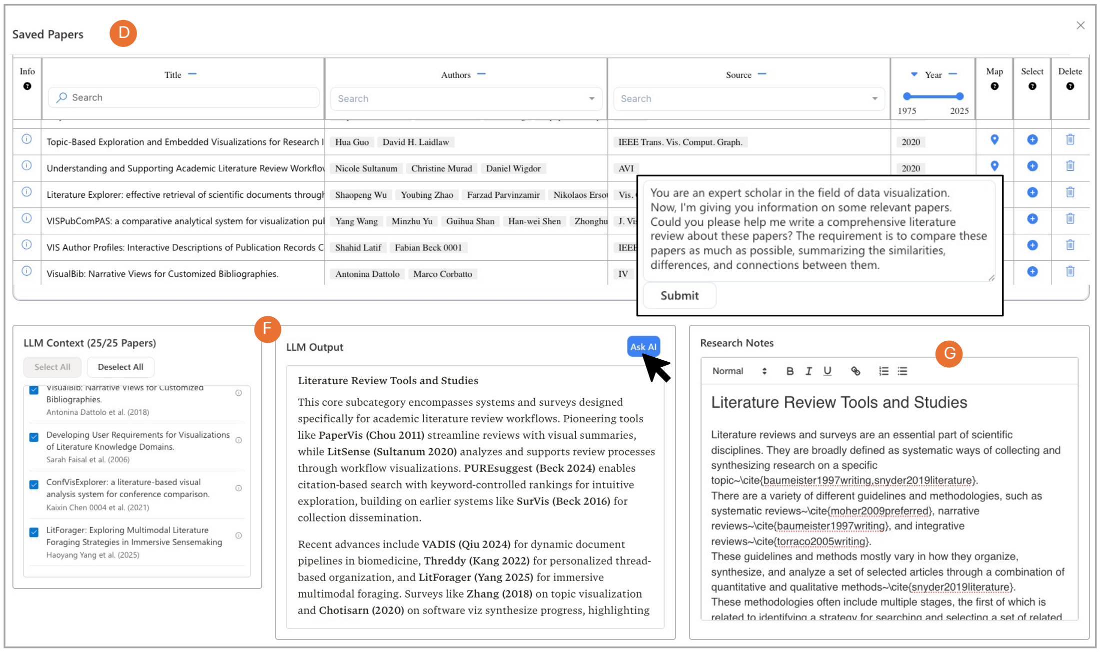

Fig. 3: The VITALITY 2 User Interface (continued). (D) **Saved Papers View** from which the saved papers can be exported as JSON. Extending VITALITY, we enabled users to generate a *Literature Review* using LLMs based on the papers that are part of the **LLM Context**, including the ability to customize the prompts. We also provide users an in-app document canvas for them to draft and organize their own literature review.

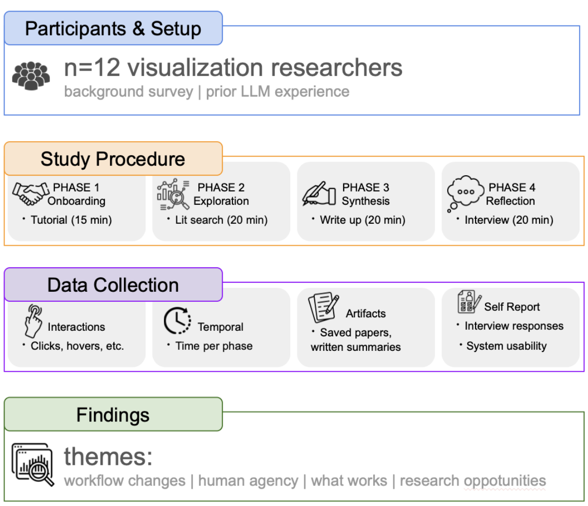

Fig. 4: Qualitative Study Procedure

4.1 Research Questions

We organize our study and subsequent analysis around two central research questions.

- **RQ 1**: How do researchers use LLM-powered literature research tools? Does this change the way of conducting literature research?
- **RQ 2**: What is still missing, especially interactive visualization features to support AI-assisted literature reserach?

4.2 Participants

We recruited $n = 12$ participants. To be eligible, participants must be graduate research students or hold a PhD degree in the area of visualization, since the dataset focuses on visualization research. To recruit participants, the authors advertised on their X accounts and through various Slack workspaces. Among the participants, 9 were identified as men, 3 as women. Seven participants were PhD students (2 in their first or second year, 5 in the third to fifth year), and the other 5 have a PhD degree in Visualization. For each participant, a £10 donation was made to RSPCA, the leading UK animal welfare charity. Sessions lasted approximately 75 minutes. The study was approved by department ethics committee (No. 20422401). We obtained informed consent from all participants before the start of the experiment by signing an online form.

4.3 Task

We recognize that there are several different use cases for literature research, depending on a user's background and the final goal: from getting a quick overview of an unfamiliar topic to identifying the topic for a future research project or writing a literature survey. For this study, we chose "identifying research gaps", because the latest literature tools are already quite capable of providing a quick overview, and preparing a systematic literature review would require much more time to complete.

Thus, participants were instructed to perform a literature research task using VITALITY 2 to elaborate on and justify a research idea they had previously considered but had not yet fully explored. The aim of their literature research was to situate their idea with existing research and identify potential research gaps through analyzing and synthesizing the collected literature. Participants were not required to come up with a completely new idea but they were encouraged to generate new approaches or hypotheses if they found any inspiration during their literature research. The final deliverable was a short report (approximately half a page) describing their idea, summarizing the most relevant research background, and an analysis of the identified research gap.

4.4 Procedure

Participants scheduled time with the study administrator to use VITALITY 2 while the administrator observed via Zoom's screen-sharing feature. After providing informed consent, participants observed a 15-minute tutorial that demonstrated the features of the system, then completed a step-through tutorial. Then, participants were instructed to choose a topic of their choice to conduct literature research (20 minutes). Next, participants were asked to synthesize a research idea from the papers they had identified with VITALITY 2 (20 minutes). Participants were instructed to enter their review text in an empty text box in the system. After completing their review, participants completed a semi-structured interviews that lasted approximately 20 minutes.

4.5 Measures

We recorded user interactions with the VITALITY 2 and conducted exploratory analysis of the resulting logs. We measured time spent in different stages of literature search (searching for papers, reading/understanding the work, synthesis/writing) and interactions across using different features of the tool (table view, UMAP, chat feature, etc). After completing the task, interview questions centered on usability of the tool, general experience with AI and literature research, and cognitive load.

5 FINDINGS

We organize our findings as follows. Sections 5.1-5.3 address **RQ 1** via exploration of workflow patterns, workflow changes, and what remains manual. Sections 5.4-5.5 address **RQ 2** by discussing what works and what is missing and opportunities for visualization research.

5.1 Workflow Patterns

Based on the provenance captured (Fig. 5), we reconstructed the time participants spent on each phase of research and using different features. On average each participant spent roughly 80% of time working **manually** ($M = 79.5\%$, $SD = 7.2\%$) and the rest was with **LLM-based** features ($M = 20.5\%$, $SD = 7.2\%$). On average participants spent 52.3% of the time **foraging** papers and 47.7% **synthesizing** them, but this varied significantly among participants ($SD = 23.2\%$). We observed several notable workflow patterns by analyzing the logs and the qualitative data:

Exploration vs. Goal-Direction. We observed both exploratory and goal-directed behaviors during the literature research task [69]. Some participants started with a clear goal. For instance, P5 started with a detailed plan: "I asked it [LLM] to first draft a literature review plan based on my idea." On the other hand, some participants may begin with a less defined idea and refine it as they explore. For instance, P7, upon seeing persona in a paper, mentioned "Oh, this is actually a better idea. If we go to [the research topic] group decision making, it would involve persona." This can also lead to a change of research direction. For instance, P9 began with a research idea centered on "uncertainty," but later found the projection problem more interesting as he explored the visualization.

Iterative vs. Sequential Foraging and Synthesis. In general, sensemaking is characterized as an iterative process [70]. Here we examined how such iterations unfolds in practice based on the collected provenance. We identified four major patterns categorized by what participants prioritized during the process and how they structured the specific sensemaking phases accordingly. The first group of participants preferred **comprehensive foraging *before* synthesis** and demonstrated a **sequential** workflow where they collected as many relevant papers as possible before entering the synthesis phase. This is exemplified by P4 (see Fig. 5B), who engaged heavily in searching before synthesizing. During the foraging stage, they filtered results, conducted similarity search, went through search results, searched again, etc., screening in total around 100 papers. As P4 said, "this gives me a feeling that I have gone through it. I feel *confident*". Other participants might prioritize **forming a rough understanding first** and then **iterate** through iterative foraging and synthesis. In this pattern, participants started with a couple of papers, synthesized them to form an initial understanding, and then refined the search scope and strategies to get more papers. For example, after established a list of several most relevant papers, P7 opened the synthesis panel (Fig. 5E) to screen the papers, formulated an understanding of related sub-problems, and went back to search.

Digging into the iterative process, we observed two more extreme categories of how the different phases were connected. The first is **synthesis *while* foraging**, i.e., the synthesis stage is **tightly coupled** with the foraging stage; they externalized their thoughts by taking notes and writing the report. This can be seen in Fig. 5 where there are instances of overlapping green and yellow rectangles. In this pattern, participants tried to digest every piece of relevant information upon seeing them. For example, P1 wrote down the potential themes to organize the papers while screening them (Fig. 5A): "I usually form a framework of what's in there as I go through the papers." The other pattern was **synthesis *before* foraging**. This was seen in the workflow of P5, who started writing and structured the report before starting search (Fig. 5C) and used LLMs to summarize and synthesis the search results before actually screening them. They only screened the paper when it occurred in the generated summary which needed verification. The difference in such patterns suggest that the preferred timing may vary when incorporating AI assistance to their workflows.

Strategies for Handling Unsuccessful Paper Retrieval. Some participants adjusted the search scope immediately, e.g., P6 commented "I don't want to synthesize using only 2 papers, so I will pivot my topic here [...] ." Others persisted in trying to identify the relevant literature by tweaking the queries or trying different search methods. They sought to ascertain whether the issue was in the framing of the query or due to a lack of relevant papers within the chosen scope. For instance, P11 spent a lot of time experimenting with different keywords, seed papers for similarity search, and using the chat window (Fig. 5F).

5.2 Workflow Changes

Improving Efficiency. Participants noted several places where LLMs improved the efficiency of their literature review process. One key improvement was in **establishing an initial set of seed papers.** In traditional literature research, it can be difficult to articulate suitable keywords for search. Participants noted that similarity search and the chat feature in VITALITY 2 made this easier. For instance, P10 said "I don't think my traditional way is efficient, because I prefer to have keywords in Google Scholar... but I need to check one by one, like, go through their abstracts one by one to see if this is relevant or not, and I personally think that's not quite efficient." P10 went on to comment "If I use the similarity search feature here, and... because I could say there's a ranking for similarity, and If I, yeah, just see the top 5 or top 10, I could see [...] They are relevant to the keywords here."

Though not all participants used chat features frequently, some participants had a strong preference of using AI to find an initial set of seed papers. P3 said, "I feel searching papers manually is inefficient. I prefer to have AI help me find them; at least to start, help me find the 3-4 most relevant papers. I feel many article titles and abstracts aren't really the same as what they actually did, or they don't fully match what I want to know." Similarly, P6 commented, "Yeah, when I make that query, I let LLM pick some papers, and I let them write the paragraph explaining why this paper is something I have to understand."

Another area of improved efficiency was in **screening and understanding papers quickly.** While reading through full paper texts, or even just abstracts, can be time consuming, participants were delighted by the help of LLMs to screen and understand papers more efficiently. P6 noted, "for most time, there is really few cases I actually need to read the original abstract or skim the paper [because of the LLM screening], unless I think it is a really important one." They also clarified that this had its limits: "all I generate with the LLM is to help me understand, and I write the interpretation note myself, and I synthesize it myself." In a similar vain, some non-native English-speaking participants, such as P4 and P6, used LLMs to translate English content into their native language for easier reading and comprehension. P6 commented, "I sometimes try to translate one by one into Korean, because I feel more comfortable reading in my mother language."

Next, participants also noted that LLMs improved the speed at which they could **get a high-level overview (of a new domain).** P9 commented, "when you ask high-level questions, like, give me an overview of those 10 papers and categorize them into what I want, ... that's pretty good, normally."

Finally, participants also noted that LLMs helped them to **generate first drafts of writing more quickly.** For instance, P12 said, "Oh yeah, [LLMs produce] a very nice draft." It helped inspire structural changes to the writing, e.g., P12 said "Oh, I want the structure, there's some context, something I have, maybe some very rough draft I have."

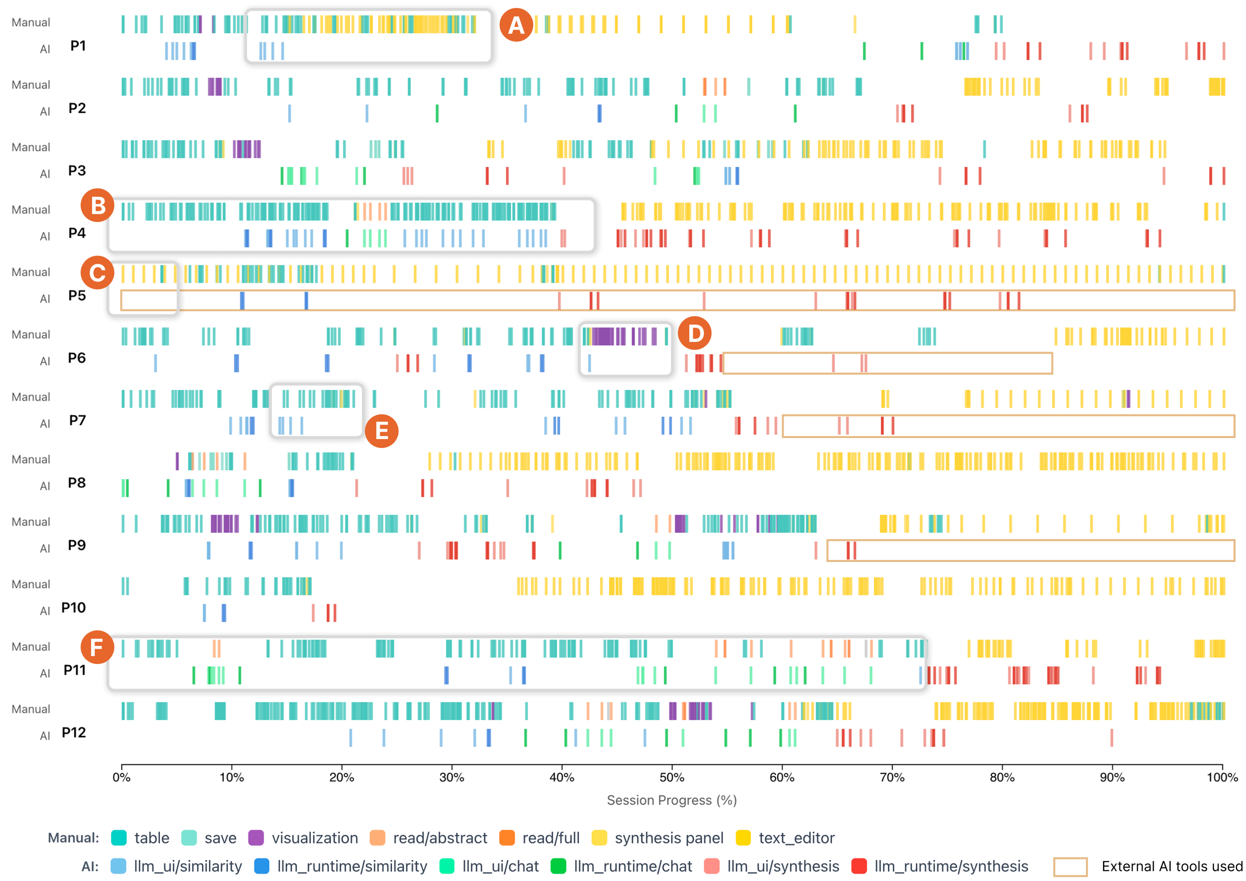

Fig. 5: Provenance logs capturing participants activities when they used VITALITY 2 to complete the literature research task. The logged interactions are categorized into *manual* strategies (e.g., table search and filtering, visualization, writing in text editor, etc.) and *AI/LLM-assisted* approaches (e.g., similarity search, chat window, and LLM synthesis).

Reducing Cognitive Load. In addition to improving efficiency, LLMs also helped reduce cognitive load. When overwhelmed, P6 said "this [long] list of titles is a bit overwhelming. I don't want to scan through every abstract one by one, so I will try [using LLMs to write a one-sentence summary of each paper]." Similarly, when stuck deciding on keywords to use, P3 asked the LLM to recommend keywords: "I'd like to search for papers about how to retrieve a specific part of an image. What keywords should I use for this type of task?" In another example P2 mentioned using LLMs to help understand unfamiliar concepts and hard-to-parse papers. Some participants also used the LLMs to suggest schemas or categorize papers, providing cognitive scaffolding. For instance, P4 prompted "help me categorize the papers based on argumentation models." When experiencing challenges in categorizing papers into dimensions, P2 used LLMs to help elicit and externalize schemas: "Sometimes I have a few categories in my mind. Then, AI might help summarize and formalize them." However, we have to note that sometimes LLMs can reduce the wrong kinds of cognitive load such as the germane cognitive load which is essential for learning [71]. For instance, P4 indicated that they sometimes "just hand everything over to AI" and "don't think for [themselves]."

Enabling Hypothesis-Driven Exploration. In their previous workflows when there were not many AI tools, participants mentioned that they needed to "go through the the abstracts [of search results] one by one" (P10) to narrow down the results to a few most relevant ones before they could digest and further synthesize the papers. However, during the study, we observed that P5 saved all the search results from a semantic query (the similarity search "by abstract" option in Fig. 2B) without screening them and directly started chatting with the papers to "get a landscape of the area" and explore more specific research questions. He explained that "during this process I'm not really searching for papers, I'm generating conclusions and then verifying whether those conclusions are correct." This suggested that LLMs might enable more efficient hypothesis-driven literature research workflows.

From Direct Human Agency to Distributed Agency. Shared among the participants, they felt that they became "managers" (P9) or "validators" (P4) during LLM-supported workflows. Some participants were happy that they could hand over tedious works to AI or LLMs and engage in what they were really interested in. Most participants commented they would do literature research for different purposes, such as exploring and justifying a research idea, or finding a technical solution to support a certain component of their system. P1 found that "with AI now, I can focus my attention on the most interesting subtopics and exploring the essential gaps; for the other parts [not the focus of research], I just have AI generate the literature review and verify them". For P5 the benefit was that "this [using ChatGPT to summarize papers] widens the range of things I'm willing to understand" which he "[didn't] particularly want to read when there was no AI assistance".

Enabling Collaborative and Conversational Knowledge Construction. While literature research was often an individual process, LLMs are transforming this to a collaborative process by acting as mentors, colleagues, challengers, etc. For example, "if unfamiliar with a field", P7 would ask AI to "be [his] mentor and introduce what has been done there". The chat interface also "provided an *conversational* environment... [this] generally make you raise questions and then think further" (P4). Even when the LLMs did not really understand the human intentions, the generated content could serve as stimuli to invoke thinking and provide inspirations, as P3 commented "it [a certain sentence in LLM-generated summary] makes a lot of sense, although I feel it's talking about something completely different from what I mentioned". In addition to having "someone" to discuss with when getting stuck, participants felt such interactive interfaces make it possible to "think explicitly with documentation" (P4). However, participants also mentioned the potential risk of treating LLMs as collaborators as "it (LLM) is really good at providing feedback, like sometimes it tells me this is a Nature-level idea, which makes me quite inflated" (P4).

5.3 Human Agency – What Remains Manual

As shown in Fig. 5, while LLM features were frequently used during the literature research process, participants still relied on manual interactions most of the time. Here we synthesize the essential processes that still require manual efforts.

Steering Research Directions and Defining Scopes. As mentioned in Section 2.1, there have been fully automated literature research tools. However, participants agreed that human expertise and judgment are still essential for steering the research direction, especially for *ill-defined problems*. When a problem direction is too generic, the results may include many irrelevant papers. For example, P2 started by searching annotation tools but realized the meaning of annotation can vary in different contexts such as machine learning, qualitative research, etc. While his actual intention was tools for annotating cultural heritage data (images and 3D models), there are fewer papers directly addressing this problem. He needed to iteratively re-define the scope to find papers that are more likely to be relevant. Also, current LLMs are effective at capturing lexical and semantic similarity, but they may struggle to identify latent relationships between papers without strategic prompting or guidance from users. For example, problems can be discussed before with different framings. P11 intended to find papers on visual interfaces for developing LLM agents but couldn't get relevant results after many attempts. Finally he tried prompting the LLMs to identify similar problems examined before and obtained an effective suggestion, i.e., searching for papers on visual programming which has been studied years ago. While LLMs were involved, the processes required human steering to identify the appropriate scope and direction.

Furthermore, participants highlighted that the research process is only *meaningful* when humans are involved. Personal interests and preferences matter, as P4 said, "AI can do that well and find interesting problems, but those interesting problems aren't yours." P5 believed "end-to-end automation would mean there's no human at the start or the end, that would make it meaningless to humans."

Interpretation and Knowledge Construction. Although LLM-powered tools enabled more top-down processes (as described in Section 5.2), it does not mean that the early stages in sensemaking can be skipped or LLMs can eliminate human efforts in understanding the papers. The terms in LLM-generated answers are sometimes "too general and not very comprehensive" (P7) and don't always make sense. P6 highlighted the difference between *summarization* and *interpretation*, "I try to write the interpretation notes for each paper...you're not gonna just summarize it in an objective manner, you somehow have to interpret so that you have a relation between that paper and your own project."

The knowledge synthesis or crystallization is eventually an **internal process**. P5 pointed out that "in research, most difficult things cannot be directly accelerated... you need to think in your mind", which theoretically means a certain level of understanding is required. For P6, there was not much that LLMs could help in that internal process, "not because they [LLMs] are not intelligent enough, but, when I'm doing that [synthesizing], there is no room for LLMs in my brain to intervene with that active thinking process."

Learning and Serendipitous Discovery. Participants also view literature research as a process of learning and discovery. The use of LLMs was not just for productivity or completing the task, "you won't really let it [AI] generate the related work section" (P7). Instead, P7 mentioned they often "treat it [AI-generated content] as a tutorial to educate oneself." For P5, an essential role of LLMs is that "based on [his] existing knowledge scope, it generates some summaries that allow [him] to incrementally expand [his] knowledge." Sometimes participants encountered papers that were not directly relevant to the ongoing task but interesting to them. For example, P2 found a few papers that were "not the type [he] was looking for but the topic was interesting." It is likely that such papers would be categorized as irrelevant and thus filtered out if the process were fully automated, or never encountered if the process were fully manual.

Humans Remain More Efficient at Certain Tasks. AI does not always work faster than human. For example, participants could instantly **tell the relevance of papers they knew before**: "because I know this paper, so I don't think that's highly relevant" (P10). Another shared view among participants was that currently it is more efficient to manually **curate a proper or thorough paper collection**, such as when writing a survey paper or related work sections, compared to using automated tools. There is no easy way to validate if those tools have followed rigorous methods (e.g., PRISMA) to collect papers or if important works are missing. Participants felt such processes *"uncontrollable"* and "it's hard to *track* the search process" (P5). As a result, the time cost of validating such processes performed by AI tools could be higher than going through a manual collection process.

5.4 What Works and What is Missing

In this section, we report participants feedback on the VITALITY 2 system, including what works and possible improvements.

Database Search vs. Online Search. The VITALITY 2 interface is built on a curated dataset consisting of academic papers from visualization and HCI literature. Currently the system does not have access to online information so the literature search is performed within the dataset. Participants generally appreciated the dataset as it provides *reliable* data sources for LLMs to retrieve so that "it will not return fake papers" (P8). Also it is more likely to get *relevant* and *high quality* results, as P1 commented "having an agent collect papers automatically doesn't really work well: it tends to grab arXiv papers and cannot access IEEE Xplore or ACM Digital Library." The dataset also provided P4 with "a sense of *finiteness* and *confidence*" as the boundary of the search results was clear. In contrast, "I always had a feeling of uncertainty and thought there was more [papers] to be discovered [when using ChatGPT to search]" (P4). However, as visualization research is often interdisciplinary, it is difficult to cover every related venues. For instance, P5 realized "model-based things [such as benchmarking agents] are not in the visualization or HCI collection." Also, there is a delay between when papers are published and added to the collection, so it is still necessary to check online for the most recent publications.

Keyword vs. Similarity vs. RAG vs. Agentic Search. Participants combined multiple search strategies during the study. Most participants started with **keyword search** to *explore* what was in the dataset. For example, P2 reflected "At first I didn't have a particular strategy; I wanted to look at the results from the keywords, then decide the next step." When participants knew the right keyword, the search result could be "pretty accurate" (P7). But in other cases, participants felt the keyword search less effective, as it was either returning too few results (P12 said "I was expecting more from the exact match of the keyword") or too many and "*overwhelming*" (P6).

Participants agreed the **similarity search** was useful. P10 found the result was accurate, "all those recommended papers are highly relevant to my research question." P10 appreciated this functionality saved his time, "if I do this manually, it might take me longer." Similarly, the results of the chat window, which integrated the **RAG** technique, were generally useful, but in some cases "the [results of] chat interface is no better than similarity search." Some participants felt the interface design somewhat redundant given the effectiveness of the similarity search. P9 suggested "hooking the AI up to the search bar, that would give you a simpler interface."

There were cases where both similarity search and RAG did not work. For example, P11 tried to find work on interfaces for *creating* LLM agents but got papers for *using* them. This might be the result that the similarity search or RAG does not capture the nuances in the query and "it can't really judge relevance" (P11). After trying the same query using Perplexity [72], P11 found "the results from [Perplexity] are much better than [VITALITY 2]." Hence, he suggested **agentic**

search might work better as the agent can "generate some keywords" and "screen what is relevant."

Tool Integration. Participants found it convenient that the search and synthesis are integrated in one system in VITALITY 2. They felt the experience was *coherent*, as P4 commented, "the process was not fragmented... there was no need to switch between different tools, so I won't forget what I was thinking and have to look back". P6 normally used a custom UI he built for himself for literature research and found VITALITY 2 "a more integrated environment where I can do many things [within it]." P5 wanted easier integration with other tools: "all intermediate results should be in formats like Markdown so I could work with them in other tools." P1 commented that "online system is user-friendly for humans but less so for AI agents [which often run in local IDEs] as it's less convenient for them to access web interfaces than local file systems".

5.5 Opportunities for Visualization Research

In this section we discuss future research directions on how visualization can further facilitate the human-AI collaborative literature research.

Improve Explainability and Interpretability of Paper Projections. As shown in Fig. 5, not all participants engaged with the Visualization Canvas. After exploring the visualization for a while (Fig. 5D), P6 felt he "had no idea what I can do with the visualization." Observing that the selected papers forming two clusters, P9 became interested in what differentiated the two groups. He then used the LLM synthesis feature in the Saved Papers view to categorize the papers and obtained the same cluster results. While the clustering result made sense, it was difficult to interpret the distribution by simply examining the visualization. After spending more time interacting with the neighborhood of selected papers, P9 found "the high-dimensional projection to 2D [space] is still not interpretable" and he didn't think "we utilize all the power of those [the neighborhood of selected points]." One potential direction he suggested was "combine the UMAP [used in VITALITY 2] and [other projection methods] to see a more structured latent space to explore."

Visualization for Validation. With the increasing usage of LLMs for literature research, P5 found "it became challenging to validate the AI-generated content" and considered visualization as "an accelerator for people verifying information." This is consistent with recent general discussion on this issue [73], and here we discuss the observations and results from our study. As mentioned in Section 5.3, one challenge when using LLMs to collect papers is that it is difficult to verify the search results and process by LLMs. To address this, P9 mentioned they typically "switch between multiple AI tools or using different prompts [to compare the results]." P1 reflected that he would "look at the time span and then at representative papers in the field" to assess the coverage of his collection. For P7, it it important to show the intermediate steps of AI workflows as "a black box can't win people's trust." Thus, we see an opportunity here to assist users in validating results through **visual analysis of agent provenance and results**. P9 pointed out that "for what AI gives us, we can do verification, but for what AI doesn't give us, there is nothing we can do." The **visualization of the latent space**, therefore, could be useful to inform "if there is anything you missed."

One way visualization may support users is by **visualizing how generated content relates to user needs** so that it is easier to "cherry-pick really valuable things from all that 'junk' [generated by AI]" (P4). What is more important is to accelerate the knowledge construction process to enable human users to judge what does not make sense to them previously, which we will discuss next.

Visualization for Knowledge Construction. Visualization improves the efficiency in consuming information in that the visual channel enables a high bandwidth to process information [74]. P1 believed that "visualization or visual diagrams [can be helpful for] showing the structure, relations, and provenance of information, [as] it is very hard to describe these in language which is linear." Specifically, P3 highlighted the need for **interactive visualization of evolution of research works in an area**. This "isn't just a simple citation relationship" and requires deep insights into the influences of the works. P6 shared a tool he built to support his literature research workflow which included a **visual workspace for sensemaking** that enabled him to organize papers and themes visually. Related to the verification challenge, as P1 put it, "the process that LLMs synthesize information is *opaque*", there is a need to **visualize the knowledge structure and provenance of knowledge construction** to better support collaborative synthesis.

6 Limitations and Future Work

As with all empirical experiments, this study is not without its limitations. First, our study is limited in scope with respect to participants, domain, and task. We had 12 participants, all of whom were visualization researchers. More participants and literature research of multiple areas could lead to more general findings. Also we focused on the task of gap analysis, which does not cover the entire literature research process, due to the time constraint of our study. Most participants agreed that they were willing to integrate VITALITY 2 into their workflow. This suggests that, in future work, VITALITY 2 can be used to examine longitudinal use of LLM-supported literature research tools to expand our findings.

Second, while VITALITY 2 has incorporated LLM-based search and synthesis features, there are emerging AI features that can be further integrated into the system, such as *agentic search* and *allowing LLMs to access online information*, suggested by the participants. This will not only improve the effectiveness of our tool but also allows future research to examine the changes or evolution of researchers' workflows as AI technique advances.

Finally, our findings reveal various patterns of LLM-assisted literature research, but currently there is no easy way to "tell the optimal workflow" (P5) given a specific task or for a certain user, such as how much automation to use. For example, early-stage researchers may benefit from the exploration or learning process [75] while others may prioritize efficiency. More empirical studies are needed to look into the details of these patterns and examine how LLM assistance influences cognitive engagement [53] and the development of cognitive skills across different phases of the literature research process. Particularly, there are opportunities for the visualization community to explore how visualization can assist in such processes by optimizing cognitive load management while keeping humans actively engaged.

7 Conclusion

We conducted a qualitative user study with N=12 experienced visualization researchers to understand (1) how researchers are using LLM-powered literature research tools and (2) what is still missing to support AI-assisted literature research. To support the study, we introduced a system, VITALITY 2, for literature review using a RAG architecture, a corpus of more than 75,464 papers from visualization-related venues, and novel features for interacting with the paper corpus through advanced LLM techniques such as chain-of-thoughts. The system is made available online, together with its source code and the paper corpus, to hopefully stimulate future work in optimizing literature review methods. Our study revealed that LLM usage permeates the stages of literature research to varying degrees. Participants found LLMs improved efficiency and reduced cognitive load for tasks like establishing a set of initial seed papers, or screening and understanding papers quickly, yet preferred to maintain human oversight over fully automated approaches for steering overall direction and maintaining a meaningful connection to the results. Our system and study contribute to the growing body of work on LLM usage in literature research, and our findings suggest several directions of future research.

Supplemental Materials

VITALITY 2 is available is available as open-source software at `https://vitality-vis.github.io`. Supplemental material is (will be) available in PCS (the IEEE Xplore digital repository) and it includes (1) a video demonstration of VITALITY 2 and (2) the protocol of our qualitative study.

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